

The compound returns of *SMB* reach a maximum right around the early 1980s—just after the early Banz and Reinganum studies were published. Since the mid-1980s there has been no premium for small stocks, adjusted for market exposure. International evidence since the mid-1980s has also been fairly weak. Examining international data, Dimson, Marsh, and Staunton (2011) state that if researchers today were uncovering the size effect, “the magnitude of the premium would not command particular attention, and would certainly not suggest there was a major ‘free lunch’ from investing in small caps.” Fama and French (2012) also find no size premiums in a comprehensive international data set over recent periods.

There are two responses to the disappearance of the size effect. First, the original discovery of the size premium could have just been data mining. Fischer Black (1993) made this comment immediately after Fama and French’s paper was released. The “discovery” of the size effect was then an example of Rosenthal’s (1979) “file drawer problem,” which is now a “hard drive problem” (and turning into a “cloud problem”). Researchers store on their hard drives 95% of the results that are statistically insignificant (using a standard p -value of 0.05 to judge significance) and only publish the 5% that are statistically significant. The discoverers of the size premium accidentally fell into the 5% category and were just lucky. One telling outcome of data mining is that an effect appears significant *in sample*, where the models are originally estimated, but it fails *out of sample*, where the models are tested after their discovery. Banz’s size effect, therefore, might never have truly existed in the first place, and its finding by Banz and Reinganum was pure luck.²²

The second response is that actually the size effect was there and actions of rational, active investors, acting on news of the finding, bid up the price of small cap stocks until the effect was removed.²³ In this context, the disappearance of the size effect represents the best of the Grossman–Stiglitz (1980) near-efficient market in which practitioners quickly exploit any anomaly. Viewed this way, size does not deserve to be a systematic factor and should be removed from the Fama–French model.

It should be noted that small stocks do have higher returns, on average, than large stocks. The effects of other factors, like value and momentum, which we discuss below, are also stronger in small stocks.²⁴ Small stocks also tend to be more illiquid than large stocks. The pure size effect refers to the possible excess returns of small stocks after adjusting for CAPM betas. The weak size effect today means that an asset owner should not tilt toward small stocks solely for higher risk-adjusted returns. There may only be a case for preferring small caps based on

²² Harvey, Liu, and Zhu (2013) examine hundreds of factors explaining stock returns and investigate the effects of data mining on identifying factors.

²³ Schwert (2003), among others, present this argument.

²⁴ See Loughran (1997) for size–value interactions and Chen, Hong, and Stein (2007) for size–momentum interactions.